

Challenge 4: Stakeholder Engagement Plan

Optimising RAG Document Retrieval for Agronomic Advice

Note: The two stakeholder groups below are selected from the project scope and Data Card. Cross-check them against your team's Course 4 stakeholder map before submitting.

1. Smallholder Farmers

Who they are: Farmers who need practical information about crop diseases, pests, nutrient deficiencies, soil management, fertilisers, and climate adaptation.

Why important: They are the primary intended community and can reveal whether retrieved information is understandable and genuinely useful.

What matters most: Accessibility, trustworthy information, practical relevance, local context, and protection from misleading advice.

Benefits: Faster access to relevant agricultural knowledge and less time searching technical resources.

Risks: Irrelevant or outdated documents could contribute to poor decisions. English-only resources may exclude farmers who primarily use Hausa, Yoruba, Igbo, Nigerian Pidgin, or other languages.

Engagement method — Participatory Design: Farmers would take part in structured sessions to describe common questions, everyday terminology, search difficulties, and examples of useful or misleading information. The team would test prototypes with these users and use their feedback to refine the retrieval experience and evaluation cases.

2. Agricultural Extension Workers and Advisers

Who they are: Professionals who provide agricultural guidance and interpret extension and technical resources for farmers.

Why important: They can assess practical relevance, evidence quality, terminology differences, and whether retrieved documents could create misleading impressions.

What matters most: Accuracy, traceability, evidence quality, consistency, practical relevance, and accountability.

Benefits: Faster discovery and organisation of useful extension materials.

Risks: Overreliance on automated retrieval, inappropriate generalisation across crops or regions, and confidence in superficially similar documents.

Engagement method — Deliberative Workshops: Extension workers would review representative queries and retrieved rankings, discuss correct and incorrect results, identify hard negatives, and agree on characteristics of genuinely relevant documents. Findings would be documented and used to refine retrieval and evaluation decisions.

3. Influence on Project Design

Feedback could identify missing question types, local terminology, crops, farming contexts, and underrepresented needs in the dataset. It could inform preprocessing, hybrid-retrieval weighting, query formulation, hard-negative testing, and how retrieved evidence is presented. Stakeholders could also identify failure cases that aggregate nDCG@5 does not explain. Their feedback would help determine whether further multilingual, local, expert, or community validation is required before deployment.

4. Engagement Process and Accountability

Engagement would occur in three stages: discovery, testing, and validation. The team would record stakeholder feedback, link important feedback to specific project decisions, and document reasons when recommendations cannot be implemented. This creates a transparent connection between participation and system design.

5. Responsible AI Contribution

Participatory Design gives intended users a direct role in shaping the system, while Deliberative Workshops add expert-informed review of relevance and risk. Together they can improve trust, representation, transparency, and accountability. This is important because a strong benchmark score should not be treated as proof of real-world readiness.

6. Long-Term Success and Community Acceptance

Continued engagement can keep the project responsive to changing agricultural needs and reduce the risk of building a technically effective but socially unsuitable system. Farmers can strengthen usefulness and inclusivity, while extension workers can strengthen evidence quality and practical accountability. Future deployment should include appropriate local, multilingual, expert, and community validation.